Rapid Revision Summary

Last-minute review · condensed from 16 papers and mark schemes, Jun 2019 – Jun 2026

THE PAPER

2 hours · 100 marks · 20–26 questions · calculator · ≈ 1.2 min per mark. Easy first, grade 8–9 last (vectors, functions, bounds, calculus, 3D trig). "Write down all the stages in your working."

ON THE FORMULAE SHEET

$S_n = n/2[2a + (n − 1)d]$ · quadratic formula · trapezium · sine and cosine rules · ½ab sin C · cone, sphere, cylinder, prism

Learn: arc and sector, circle theorems, index laws, bounds, nth term, SOH CAH TOA.

TOP 12 MARK-LOSERS

- "Working required" + answer only = **0**.
- Wrong answer with no working = 0.
- Simplifying a correct answer wrongly (do not ISW).
- "(1 − 4%)" instead of **0.96**.
- Proportion without **k**.
- Trial and improvement where algebra is required.
- Wrong bound chosen for a − b or a/b.
- 35×10^{-138} is **not** standard form.
- Circle-theorem angle with no reason in words.
- Vector ratio guessed, no two-route method.
- Straight segments instead of a smooth curve.
- Wrong s.f./d.p.; rounding in the middle.

NUMBER

INDICES

$a^m \times a^n = a^{m+n}$ · $a^m \div a^n = a^{m-n}$ · $(a^m)^n = a^{mn}$ · $a^0 = 1$ · $a^{-n} = 1/a^n$ · $a^{m/n} = (\sqrt[n]{a})^m$

$(5a^4r^2)^3 = 125a^{12}r^6$ · $2 \times 16 = 2^5$

- Standard form $a \times 10^n$, $1 \le a < 10$: $(5 \times 10^{42})(7 \times 10^{-180}) = 3.5 \times 10^{-137}$.
- Surds: $\sqrt{675} = 15\sqrt{3}$; $21/(3 - \sqrt{2}) \times (3 + \sqrt{2})/(3 + \sqrt{2}) = 9 + 3\sqrt{2}$.
- HCF lowest powers, LCM highest powers.
- $x = 0.25454\ldots$: $1000x - 10x = 252 \Rightarrow x = 14/55$.

BOUNDS

Nearest 10 → ±5 · 1 d.p. → ±0.05 · 2 s.f. → half the last unit.

Max (a − b) = $a_{max} - b_{min}$ · max (a/b) = a_{max}/b_{min}. (9.65 − 3.75)/1.835 = 3.22. "Suitable accuracy": both bounds round to the same value.

PERCENTAGES AND PROPORTION

- Reverse: 198 ÷ 0.72 = 275.
- Repeated: 750 000 × 0.96 × 0.935 = 673 200; 698 445 ÷ 673 200 = 1.0375 ⇒ 3.75%.
- Unknown rate: divide out known years, root the rest.
- $y \propto 1/x^3$: $y = k/x^3$, $k = 4 \times 27 = 108$; $y = 864 \Rightarrow x = 0.5$.
- Ratios: 7 : 2 and 5 : 9 → 35 : 10 : 18.
- $s = d/t$ · $\rho = m/V$ · $p = F/A$ (convert units first).

ALGEBRA

- $5x(x + 2)(3x - 4) = 15x^3 + 10x^2 - 40x$.
- $50g^2 - 18 = 2(5g - 3)(5g + 3)$.
- $(4x^2 - 9)/(2x^2 - 5x - 12) = (2x - 3)/(x - 4)$.
- $5/4 + (x - 3)/6x = (17x - 6)/12x$.
- Subject twice: collect, factorise, divide.
- $7 + 12x - 3x^2 = 19 - 3(x - 2)^2$ → max (2, 19). $5x^2 - 20x + 23 = 5(x - 2)^2 + 3$.

NON-LINEAR SIMULTANEOUS

$y = x + 2$ into $x^2 + 3y + y^2 = 7 \Rightarrow 2x^2 + 7x + 3 = 0 \Rightarrow$ (−½, 3/2), (−3, −1).

- $2x^2 + x - 28 > 0 \Rightarrow (2x - 7)(x + 4) > 0 \Rightarrow x < -4$ or $x > 3.5$.
- $-4 \le 2y < 6$, y integer: −2, −1, 0, 1, 2. Dashed lines for < and >.

SEQUENCES AND FUNCTIONS

- $u_n = a + (n - 1)d$. $a + 19d = 123$, $2a + 29d = 192 \Rightarrow d = 6$, $a = 9$.
- Terms 50 to 100: $S_{100} - S_{49}$.
- fg(x): g first. $f = 9 - \sqrt{x}$, $g = 4x^2$: $9 - 2x < 0 \Rightarrow x > 4.5$.
- Inverse: $3x/(2 - x) \rightarrow 2x/(3 + x)$. Exclude x making a denominator 0.

GRAPHS AND COORDINATES

- Perpendicular gradients: $m_1m_2 = -1$. Line form ay = bx + c with integers.
- Tangent to circle: ⊥ radius. P(8, −10): tangent gradient 4/5.
- f(x + a) left a · f(x) + b up b · f(ax) x ÷ a · −f(x) reflect in x-axis. (8, −12) → f(2x): (4, −12).
- Plot within ½ square; smooth curve.

CALCULUS

- $d/dx(ax^n) = nax^{n-1}$. Turning point: dy/dx = 0.
- $y = x^3 - 40x + 1$, gradient 8 ⇒ x = ±4 ⇒ (4, −95), (−4, 97).
- $V = 3x^2(13.5 - 2x) \Rightarrow dV/dx = 81x - 18x^2 = 0 \Rightarrow x = 4.5$, V = 273.375.

- v = ds/dt, a = dv/dt. Tangent at a minimum: y = minimum value.

GEOMETRY AND TRIGONOMETRY

- Cosine rule then sine rule: BD = 13.3 → BC = 10.4.
- Area ½ab sin C. Bearings clockwise from north.
- 3D: find the right-angled triangle; base lengths first.
- Exterior angle 360/n: 18° ⇒ n = 20.

CIRCLE THEOREMS

Centre = 2 × circumference · semicircle 90° · same segment equal · cyclic quad 180° · tangent ⊥ radius · equal tangents · alternate segment.

MENSURATION AND SIMILARITY

- Arc $(\theta/360)2\pi r$ · sector $(\theta/360)\pi r^2$ · segment = sector − $\frac{1}{2}r^2 \sin\theta$. 140°, r = 16 ⇒ 313 cm^2.
- Hemisphere: ½ sphere (+ πr^2 flat face). Frustum: big − small.
- Scale k: area k^2, volume k^3. 10 : 15 ⇒ 8 : 27; 19 parts = 1197 ⇒ A = 504.

VECTORS

METHOD

Two routes to one point; scalar λ/n; equate a and b coefficients.

PQ = −6a + nb = λ(−30a + 24b) ⇒ λ = 1/5, n = 4.8.

STATISTICS

- Grouped mean: Σ(mid × f) ÷ Σf = 920 ÷ 100 = 9.2. Modal class = interval.
- Cumulative frequency: upper bounds; median n/2; IQR = UQ − LQ.
- Frequency density = f ÷ width; frequency = area. 22/126 = 11/63.
- Combined means: totals = mean × n.

PROBABILITY AND SETS

- Multiply along, add between. Without replacement: denominators fall.
- Exactly two same (9R, 7Y, 4G, take 3) = 3 × [RRx + YYx + GGx] = 51/76.
- At least one = 1 − none. Expected = p × n.
- ∩ both · ∪ either · A′ not A · ⊂ subset; fill Venn from the centre.